NONLINEAR FIT

CHEAP SCREEN

NONLINEAR FITS

INPUT
Noisy samples
Time step + terms

INITIAL FIT
Starting equations
Denoise + fit

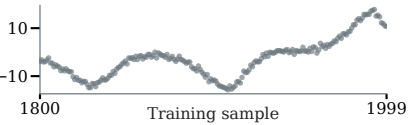
PROFILE & SCREEN
Adjust trajectory locally
Rank term combinations

REFIT & SELECT
Top 10 per round
Fit + complexity cost

stop

FINAL MODEL
Refit all samples
Selected equations

Recorded input: Lorenz x



Use the best fit
At most two rounds

Discovered Lorenz equations

$$\dot{x} = -10.08x + 10.11y$$

$$\dot{y} = 27.84x - 0.99y - 0.99xz$$

$$\dot{z} = -2.67z + 1.00xy$$